AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

SCHOOL OF COMPUTING CHENNAI

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

23CSE304 Embedded Systems Laboratory

Activity Sheet #7

Activity Name: PWM Module Interfacing

Name of Student:	**Chetan Kumar G**
Registration No.:	**CH.SC.U4CSE24109**
Activity Date:	**02|10|2026**
Submission Date:	**02|10|2026**

Assessment Rubrics:

Component	Maximum Marks	Marks Allotted
Program	3	
Output	4	
Viva	2	
Timely Submission	1	
Total	**10**	

Faculty In-charge

Write an embedded C program for ARM architecture to drive PWM module. Debug and verify the program using IDE on a development computer system.

PROGRAM:

```
#include <LPC21xx.H>

int main(void)
{
    PINSEL0 = (1 << 1);        // Select PWM1 on P0.0

    PWMPR = 0;                 // Prescaler

    PWMMR0 = 1000;             // PWM period

    PWMMCR = (1 << 1);         // Reset PWMTC on PWMMR0 match

    PWMMR1 = 500;              // 50% duty cycle

    PWMLER = (1 << 0) | (1 << 1);   // Load MR0 and MR1

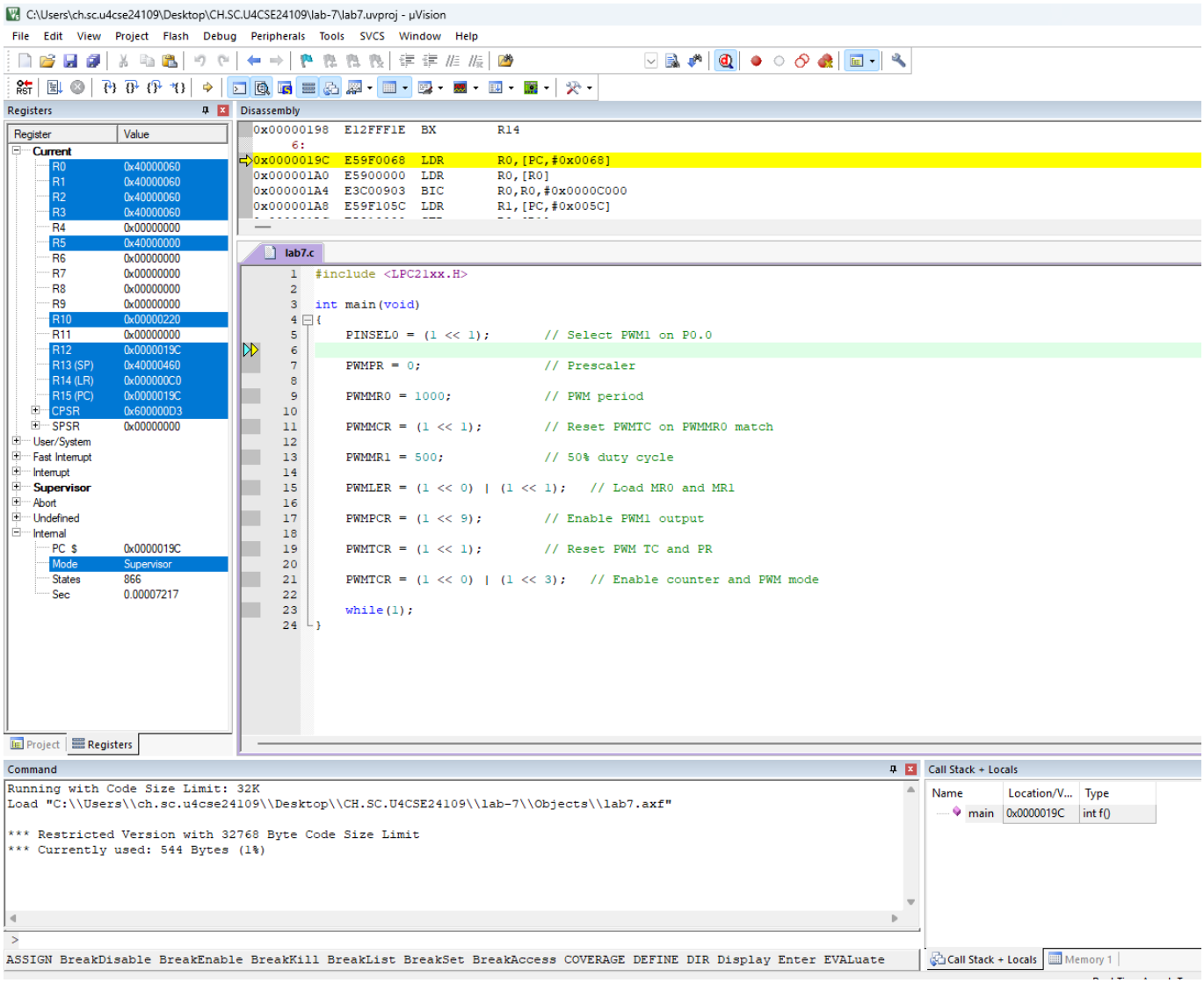
    PWMPCR = (1 << 9);         // Enable PWM1 output

    PWMTCR = (1 << 1);         // Reset PWM TC and PR

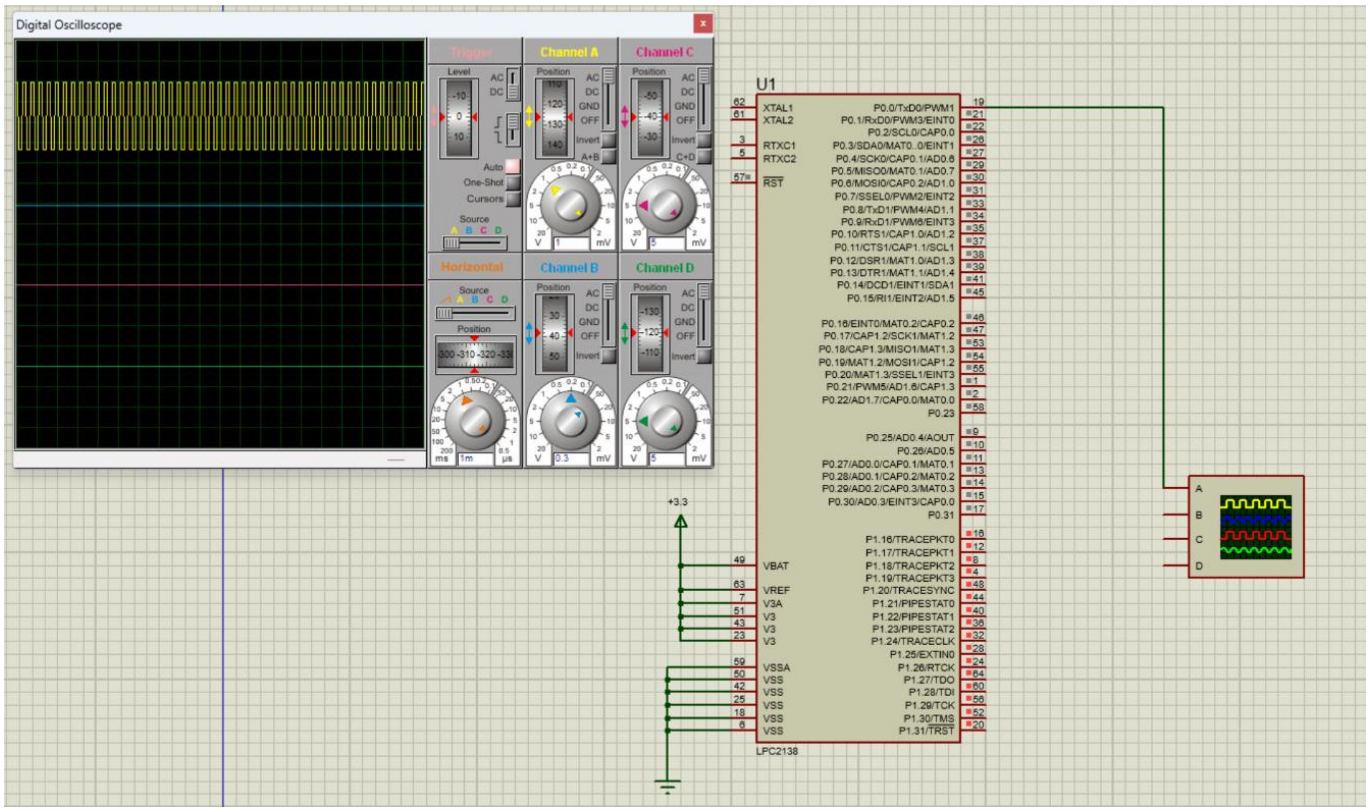
    PWMTCR = (1 << 0) | (1 << 3);   // Enable counter and PWM
mode

    while(1);
}
```

OUTPUT WINDOW:



OUTPUT:


Digital Oscilloscope
Trigger
Channel A
Channel C
Horizontal
Channel B
Channel D
U1
XTAL1
XTAL2
RTXC1
RTXC2
RST
VBAT
VREF
V3A
V3
VSSA
VSS
LPC2138
+3.3
A
B
C
D